

X-CUBE-STL software and documentation errata

Introduction

This document applies to the X-CUBE-STL STM32Cube Expansion Package. The X-CUBE-STL references concerned by the errata in this document are summarized in Table 1 together with the related Arm® processors.

Table 1. Impacted variant summary

X-CUBE-STL reference ⁽¹⁾	Arm® processor
X-CUBE-STL-F3	Cortex®-M4
X-CUBE-STL-F4	
X-CUBE-STL-G4	
X-CUBE-STL-H7DC (Cortex®-M4)	
X-CUBE-STL-L4	
X-CUBE-STL-MP1 (Cortex®-M4)	
X-CUBE-STL-WL	
X-CUBE-STL-F7	Cortex®-M7
X-CUBE-STL-H7	
X-CUBE-STL-H7DC (Cortex®-M7)	
X-CUBE-STL-L5	Cortex®-M33
X-CUBE-STL-U5	

1. Refer to each erratum for the references and versions considered.





1 General information

The X-CUBE-STL STM32Cube Expansion Package runs on STM32 microcontrollers and microprocessors based on the Arm® Cortex® processor.

Table 2 defines the acronyms needed for a better understanding of this document.

Table 2. List of acronyms

Acronym	Description
API	Application programming interface
DC	Diagnostic coverage
DCache	Arm® Cortex®-M7 data cache
STL	Self-test library
TM	Test module

Note: Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere.

arm



2 Summary of errata

Table 3 gives a reference to the software errata and their status:

- A = workaround available
- N = no workaround available
- P = partial workaround available

An erratum workaround is deemed partial if

- either it only reduces the rate of occurrence, or the consequences, or both
- or if it is effective only for a subset of applicable references or versions
- or it is a combination of both

Table 3. Summary of software errata

Domain	Section	Erratum	Status
CPU	3.1	The CPU TM9 function may return a wrong test failure status	A
RAM	3.2	Risk of data cache and RAM misalignment during the RAM TM execution when the data cache is enabled	A

Table 4 gives a reference to the documentation errata.

Table 4. Summary of documentation errata

Domain	Section	Erratum
Flash memory	4.1	FLASH_SM_9 in specific X-CUBE-STL safety manuals



3 Description of software errata

3.1 The CPU TM9 function may return a wrong test failure status

3.1.1 Applicability

Table 5. Applicability of the CPU TM9 erratum

X-CUBE-STL reference	X-CUBE-STL versions	Arm® processor
X-CUBE-STL-F3	Up to v1.1.0 included	Cortex®-M4
X-CUBE-STL-F4	Up to v1.1.0 included	
X-CUBE-STL-G4	Up to v1.2.0 included	
X-CUBE-STL-H7DC (Cortex®-M4)	Up to v1.0.0 included	
X-CUBE-STL-L4	Up to v1.2.1 included	
X-CUBE-STL-MP1 (Cortex®-M4)	Up to v1.0.0 included	
X-CUBE-STL-WL	Up to v1.1.0 included	
X-CUBE-STL-F7	Up to v1.2.0 included	Cortex®-M7
X-CUBE-STL-H7	Up to v1.2.0 included	
X-CUBE-STL-H7DC (Cortex®-M7)	Up to v1.0.0 included	
X-CUBE-STL-L5	Up to v1.0.0 included	Cortex®-M33
X-CUBE-STL-U5	Up to v1.0.0 included	

Note: There is no CPU TM9 test module in the X-CUBE-STL release for the STM32 microcontrollers based on the Cortex®-M0, Cortex®-M0+, and Cortex®-M3 processors.

3.1.2 Erratum details

3.1.2.1 Description

The CPU TM9 function may return the STL_FAILED value even when the test is executed correctly on a nonfaulty hardware. In this case, the test result is a false positive.

Inside the CPU TM9, some internal computations depend on the GE bits from the xPSR register:

- If one of the GE bits, GE[0:3] (APSR[19:16]), is set to 1 before CPU TM9 execution (or set by an interrupt handler during CPU TM9 execution), the test result is set to STL_FAILED, false positive. Furthermore, the original GE[0:3] values are overwritten to 0, when exiting the CPU TM9 function.
- If the GE bits are set to 0, the test result is set to STL_PASSED. No abnormal behavior is noticed.

Note: The above CPU TM9 behavior is deterministic.

3.1.2.2 Workaround

Apply in the application software one of the following measures just before calling the CPU TM9 function, whatever the test mode, single or multiple test. The choice between the two measures depends on the application software, whether it relies on GE bits values or not.

- Either simply reset the GE bits from xPSR
- Or save the xPSR, reset the GE bits from xPSR, and restore the xPSR value after the test execution

Note: If an interrupt is reached during CPU TM9 execution, and if the interrupt handler sets the GE bits inside xPSR, it is still possible to get the false positive outcome from CPU TM9 (STL_FAILED) and GE[0:3] values overwritten to 0.



3.1.2.3

Environment issue description

The issue happens only when the two following conditions are met:

- At least one instruction from SADD8, SADD16, SSUB8, SSUB16, SSAX, SASX, UADD8, UADD16, USUB8, USUB16, USAX, and UASX is used inside the application software (outside STL own functions)
- The result of these instructions sets to 1 one or multiple GE bits, just before the CPU TM9 function call

If these instructions are not directly programmed by using corresponding intrinsic functions (such as __UADD8() or others) or in assembler (inline or in assembler code), their usage may depend also on the compiler and potentially on the compiler version. Note that the instructions can possibly be used in specific libraries included in the final project.

3.1.3

Functional safety implications

Claims related to the achievable DC included in the related X-CUBE-STL safety manuals are not affected by this limitation. DC values are confirmed because of the false positive outcome.

The end users must evaluate the severity of the potential perturbations of their systems caused by the GE[0:3] value change. In any case, note that when the incorrect behavior is produced, a failed test result is always produced for CPU TM9 (STL_FAILED). In principle, the recovery actions as requested by CoU_STL_3 must be initiated.

3.2

Risk of data cache and RAM misalignment during the RAM TM execution when the data cache is enabled

3.2.1

Applicability

Table 6. Applicability of the RAM TM erratum

X-CUBE-STL reference	X-CUBE-STL versions	Arm® processor
X-CUBE-STL-F7	Up to v1.2.0 included	Cortex®-M7
X-CUBE-STL-H7	Up to v1.2.0 included	
X-CUBE-STL-H7DC (Cortex®-M7)	Up to v1.0.0 included	

Note:

The RAM test module erratum applies to the Cortex®-M7 only, when the data cache is enabled by the application.

3.2.2

Erratum details

3.2.2.1

Description

The issue described below only happens if the data cache (DCache) is enabled on the product. If the DCache is not enabled, there is no issue and the workaround is not needed.

The DCache is disabled and re-enabled during RAM test execution by specific STL internal utilities to ensure that the RAM test is really executed on the RAM and not on the DCache. This is explained in the corresponding user guide in *Chapter 4.3.11* (for single core) and *Chapter 4.3.12* (for dual core).

If an interrupt handler that uses cacheable data appears at a specific timing during the execution of these DCache disable and enable utilities, a RAM/DCache misalignment may occur and wrong cacheable data (such as variables, stack, and others) can be used by the interrupt handler.

X-CUBE-STL provides no notification to detect if the issue did happen.



3.2.2.2

Workaround

The main principle of the workaround is to prevent any interrupt handling when executing the disable and re-enable DCache flow around the RAM test execution. To do so, the application must use exclusively the RAM single API function call (STL_SCH_RunRamTM()), and follow the mandatory steps below while performing the RAM test:

- disable the DCache before running the RAM test module
- enable the DCache just after returning from the RAM TM execution
- encapsulate both disable DCache and enable DCache function calls into a disabled interrupt window

Workaround proposal with the use of CMSIS procedures calls (to be adapted to the user's application):

```
STL_UTIL_Disable_IT(); /* be sure that flag STL_ENABLE_IT is not activated to really disable/
enable interrupt with the use of STL functions, or use __disable_irq(); or...*/
SCB_DisableDCache();
STL_UTIL_Enable_IT();

yyyyy = STL_SCH_RunRamTM(xxxxxx);

STL_UTIL_Disable_IT();
SCB_EnableDCache();
STL_UTIL_Enable_IT();
```

Important: *If STL dedicated functions for disabling and enabling interrupts are used (STL_UTIL_Disable_IT() and STL_UTIL_Enable_IT()), be sure that the flag STL_ENABLE_IT is not activated. If the flag STL_ENABLE_IT is activated, functions like intrinsic __disable_irq() must be used instead.*

STL performance and execution impact

To ensure that the workaround is effective, the RAM TM execution cannot be enabled anymore in multiple testing (because of the affected STL utilities); The RAM TM must be executed via single testing calls exclusively with the workaround applied.

The use of the workaround has no impact on the execution timing of the STL_SCH_RunRamTM() function. Without the workaround, the main part of the function is already executed with DCache disabled. With the workaround, only a few additional instructions are executed with DCache disabled.

The use of the workaround implies that the execution of the interrupt handler happening during the run of RAM TM is executed with the DCache disabled.

3.2.2.3

Environment issue description

The issue may occur only for the X-CUBE-STL running on the Cortex®-M7 during the RAM test when DCache is activated. Because this issue is related to a critical run with intervening interrupts, it can be difficult to reproduce it in a deterministic way.

3.2.3

Functional safety implications

Claims related to the achievable DC included in the related X-CUBE-STL safety manuals are not affected by this limitation. DC values are confirmed.

The consequence of having wrong cacheable data (variables, stack) during interrupt handler execution must be carefully assessed by the end user. The application of the proposed workaround is anyway highly recommended if the end-user system is exposed to the described issue.



4 Description of documentation errata

4.1 FLASH_SM_9 in specific X-CUBE-STL safety manuals

4.1.1 Applicability

This erratum applies to the X-CUBE-STL safety manual releases listed in [Table 7](#).

Table 7. Documents affected by the FLASH_SM_9 erratum

X-CUBE-STL reference	Document	Revision list - reference document name ⁽¹⁾
X-CUBE-STL-G4	UM2539 revision 2	STL_G4_SM_cx641301_2.0.pdf
X-CUBE-STL-H7	UM2599 revision 2	STL_H7_SM_cx633352_2.0.pdf
X-CUBE-STL-H7	UM2599 revision 3	STL_H7_SM_cx633352_3.0.pdf
X-CUBE-STL-L4	UM2475 revision 6	STL_L4_SM_cx609803_6.0.pdf
X-CUBE-STL-L5	UM2768 revision 1	STL_L5_SM_cx682428_1.0.pdf
X-CUBE-STL-H7DC	UM2844 revision 2	STL_H7DC_SM_cx710091_2.0.pdf

1. The revision list is the official document that traces the certified software and documentation baseline. For the reader's convenience, this column reports the actual document names used this list.

4.1.2 Erratum details

This erratum is related to the two sentences, indicated below as (A) and (B) in [Table 8](#), that must be replaced in the affected safety manuals.

Table 8. FLASH_SM_9-related sentence changes

Sentence		Location and update
(A)	Location ⁽¹⁾	In note 3 (related to Tested module = Flash memory) on Table 4 . <i>Diagnostic coverage</i> in section 3.2.
	Original sentence	3. Diagnostic coverage achieved by the implementation of FLASH_SM_9 in combination with FLASH_SM_7 (refer to [1] and to Table 3 for details).
	Must be replaced with	3. Diagnostic coverage achieved by the implementation of FLASH_SM_0 (refer to [1] and to Table 3 for details).
(B)	Location ⁽²⁾	In table 7 . on section 5.2, the second bullet in CoU_STL_8 text description.
	Original sentence	Flash memory test over the Flash memory sections selected to be tested for the implementation of FLASH_SM_9.
	Must be replaced with	Flash memory test over the Flash memory sections defined by the end user. ⁽³⁾

1. Refers to note 6 instead of note 3 in the safety manual UM2844. The replacement of the text is identical.
 2. Refers to the third bullet instead of the second one in the safety manual UM2844. The replacement of the text is identical.
 3. The note associated to the original sentence is still valid.

4.1.3 Functional safety implications

The sentences are reworded to remove the references to hardware-based methods from the X-CUBE-STL safety manuals, since they are not in the scope of the X-CUBE-STL certification. The claims related to the combination between FLASH_SM_7 and FLASH_SM_9 are meant to be moved to the MCU safety manual, which is the most appropriate document. Refer to the related most updated versions for details.



Revision history

Table 9. Document revision history

Date	Revision	Changes
20-Dec-2021	1	Initial release.
14-Jan-2022	2	Added the X-CUBE-STL-F3 software variant to Table 1. Impacted variant summary and Table 5. Applicability of the CPU TM9 erratum. Added Section 1 General information with Table 2. List of acronyms. Updated the software erratum description in Section 3.2.2.1 , and the software errata workarounds in Section 3.1.2.2 and Section 3.2.2.2 . Updated the documentation erratum details in Section 4.1.2 .



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